CASE STUDY REPORT

TITLE

Engineering College: Migration from Proprietary Software to Open-Source Solutions

Maharshi Vachhani [92400120195]
Deep Fachara [92400120321]
Mahipalsinh Jadeja [92400120409]

DEPARTMENT

COMPUTER ENGINEERING

ACADEMIC YEAR

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SEMESTER

5TH

SUBJECT

OPEN SOURCE TECHNOLOGIES (01CE0526)

CLASS

5 EV4

BATCH

A

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RUBRICS FOR CASE STUDY

Sr. No.	Assessment Criteria	Excellent	Good	Satisfactory	Poor	Marks
1	**Problem Analysis, Cloud Concepts & SDG Mapping**	**4 Marks:** Clear problem identification, appropriate cloud concepts, and accurate SDG mapping.	**3 Marks:** Good analysis with minor gaps in concepts or SDG mapping.	**1–2 Marks:** Basic problem understanding with partial cloud concepts or SDG mapping.	**0 Marks:** Inadequate or incorrect analysis and no SDG mapping.	**4**
2	**Security Analysis & Technical Depth**	**4 Marks:** Comprehensive security analysis, well-defined architecture, appropriate technologies, and justified solutions.	**3 Marks:** Good technical explanation with minor omissions.	**1–2 Marks:** Limited security analysis and technical details.	**0 Marks:** No meaningful security analysis or technical explanation.	**4**
3	**Report Quality & References**	**4 Marks:** Well-structured report in IEEE format with comprehensive technical content, proper citations, and relevant references.	**3 Marks:** Good report structure with minor formatting or referencing issues.	**1–2 Marks:** Basic report with limited technical details or incomplete references.	**0 Marks:** Incomplete report with poor documentation and missing references.	**4**
	Total					**12**

Sr. No.	Case Study Title / Problem Statement	COs	Date	R1(4)	R2(4)	R3(4)	Marks (12)	Signature
1	Software Development Company: Git-Based Collaborative Software Development Workflow	CO2: Version control tools, collaboration						

Notes / Instructions

1. The case study must be completed in a **group of maximum 3 students**.
2. Students must follow the **given submission deadlines**.
3. The case study report must follow the **provided IEEE Conference Paper Template format.**
4. The final report must be uploaded in **DOCX format on Google Classroom,** and a **hard copy must be submitted** as discussed.
5. The report should include proper **technical analysis, cloud concepts, security aspects, SDG mapping, figures/tables, and IEEE-style references**.
6. Use only **authentic research papers and reliable sources** for references.
7. All group members must contribute equally and be prepared for **presentation and evaluation**.
8. Remove all template instructions/content before final submission.
9. Plagiarism or copied content will not be accepted.

Ms. Priyanka Mangi
Subject Faculty

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Migrating a College Project from Proprietary Software to Open Source

Course Name and Course Name: OPEN SOURCE TECHNOLOGIES (01CE0526)

Co's Mapped – CO1,CO5

SDG's Mapped - SDG 4: Quality Education
SDG 9: Industry, Innovation and Infrastructure

Abstract
Keywords

I. Introduction
- Background of the selected case study
- Importance of Open Source Technologies in the selected domain
- Objectives of the case study
- Scope of the study

II. Organization/Application Background
- Overview of the organization/application/system
- Existing software ecosystem
- Business requirements
- Need for adopting Open Source Technologies

III. Problem Analysis
- Problem identification
- Existing challenges
- Engineering requirements
- Functional and non-functional requirements
- Constraints and assumptions

IV. Open Source Solution Analysis
- Selected Open Source technologies/frameworks
- Justification for technology selection
- Open Source license used (GPL, MIT, Apache, BSD, etc.)
- Comparison with proprietary alternatives
- Benefits of adopting OSS

V. System Architecture and Technology Stack
- Overall architecture diagram
- Technology stack
- Version Control System (Git/GitHub/GitLab)
- Web framework (Laravel/Django/Node.js)
- Database (MySQL/PostgreSQL/MongoDB)
- Web Server (Apache/Nginx)
- Development and deployment workflow

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VI. Database and Deployment Analysis

- Database design
- CRUD operations
- Database architecture
- Deployment strategy
- Reverse Proxy / Load Balancing (if applicable)

VII. Software Quality and Testing

- Testing strategy
- Unit Testing (PyTest/PHPUnit/JUnit)
- API Testing (Postman)
- Web Testing (Selenium)
- Load Testing (JMeter)
- Security Testing (OWASP ZAP)
- Testing outcomes and observations

VIII. Open Source Community and Licensing Analysis

- Open Source licensing model
- Compliance with license terms
- Community support
- Documentation quality
- Version management
- Contribution opportunities

IX. Technical Analysis

- Working of the proposed solution
- Advantages
- Limitations
- Scalability
- Reliability
- Maintainability
- Performance analysis

X. SDG Mapping and Justification

- Identify the relevant Sustainable Development Goal(s)
- Explain how the proposed Open Source solution contributes towards achieving the selected SDGs
- Justification with real-world impact

XI. Challenges and Recommendations

- Technical challenges
- Security challenges
- Deployment challenges
- Future improvements
- Recommendations for better implementation

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XII. Conclusion

- Summary of the case study
- Key technical findings
- Learning outcomes
- Industrial relevance of Open Source Technologies

References

- IEEE Reference Format
- Minimum 5–8 authentic references
- Research papers
- Official Documentation
- GitHub Documentation
- Linux Foundation Documentation
- Books

Appendix (Optional)

- System Architecture Diagram
- Database Schema
- Git Workflow Diagram
- Screenshots
- Test Reports
- Deployment Screenshots
- Configuration Files
- Additional Tables

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Engineering College: Migration from Proprietary Software to Open-Source Solutions

Maharshi Vachhani
Department of Computer Science and Engineering
Marwadi University
Rajkot, Gujarat, India
maharshi.vachhani127512@marwadiuniversity.ac.in

Deep Fachara
Department of Computer Science and Engineering
Marwadi University
Rajkot, Gujarat, India
deep.fachara128567@marwadiuniversity.ac.in

Mahipalsinh Jadeja
Department of Computer Science and Engineering
Marwadi University
Rajkot, Gujarat, India
mahipalsinh.jadeja129430@marwadiuniversity.ac.in

Abstract**—** ABC Engineering College currently depends on several proprietary software solutions for academic, project development and communication activities. The existing environment includes Microsoft Office, GitHub Enterprise, Microsoft SQL Server, Adobe Photoshop and Zoom. Although these tools provide useful features, their licensing and subscription costs can increase the overall software expenditure of the institution.

This case study proposes a migration strategy from selected proprietary software to open-source alternatives. Microsoft Office can be replaced with LibreOffice, GitHub Enterprise with GitLab Community Edition, SQL Server with PostgreSQL, Adobe Photoshop with GIMP, and Zoom with Jitsi Meet. Linux is also considered as an open-source operating-system platform for suitable college systems.

The proposed migration focuses on cost reduction, compatibility, user training, data migration, security, licensing and productivity. Instead of replacing all software simultaneously, a phased migration approach consisting of assessment, pilot deployment, testing, training, migration and monitoring is proposed. A cost-benefit analysis is also presented to evaluate the expected benefits and challenges of adopting open-source solutions.

Keywords**—** Open Source Software, OSS Migration, LibreOffice, GitLab, PostgreSQL, GIMP, Jitsi Meet, Linux, Software Migration, Engineering College.

I. INTRODUCTION

A. Background of the Selected Case Study

Modern engineering colleges depend on software for documentation, project development, database management, image editing, online meetings and collaboration.

ABC Engineering College currently uses several proprietary software products:

- Microsoft Office for documents, spreadsheets and presentations.
- GitHub Enterprise for project repositories and collaboration.
- Microsoft SQL Server for database management.
- Adobe Photoshop for image editing and graphics.
- Zoom for online meetings and communication.

The increasing cost of software licenses and subscriptions has encouraged the college management to investigate open-source alternatives.

Open-source software provides access to software whose source code and usage rights are governed by specific open-source licenses. Depending on the software, institutions can use, modify, distribute or customize it according to the applicable license.

The objective is not simply to replace commercial products but to determine whether suitable open-source alternatives can satisfy the college's technical and academic requirements.

B. Problem Statement

The college currently depends on multiple proprietary software products, resulting in recurring licensing or subscription costs and dependency on individual vendors.

The problem can therefore be stated as:

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How can ABC Engineering College migrate from selected proprietary software to suitable open-source alternatives while maintaining productivity, compatibility, security and reliability?

The migration must be carefully planned because students, faculty and administrators are already familiar with the existing software.

C. Objectives of the Case Study

The major objectives of this case study are to:

1. Identify the proprietary software currently used by ABC Engineering College.
2. Identify suitable open-source alternatives.
3. Compare the existing and proposed software in terms of cost, features and compatibility.
4. Analyze the advantages and limitations of open-source adoption.
5. Develop a phased migration roadmap.
6. Identify technical and organizational challenges.
7. Prepare a cost-benefit analysis.
8. Recommend security and licensing practices.
9. Minimize disruption to academic activities.
10. Increase awareness and use of open-source technologies.

D. Scope of the Study

The study covers migration of the following software:

- Microsoft Office → LibreOffice
- GitHub Enterprise → GitLab Community Edition
- SQL Server → PostgreSQL
- Adobe Photoshop → GIMP
- Zoom → Jitsi Meet

The study also considers Linux as a possible open-source operating-system platform for suitable college computers.

The migration strategy covers software assessment, compatibility testing, data migration, user training, pilot deployment, full deployment and post-migration monitoring.

II. ORGANIZATION BACKGROUND

A. ABC Engineering College

ABC Engineering College is considered as a representative engineering institution that provides computer-based education, student project development and online academic activities.

The college has:

- Computer laboratories
- Faculty systems
- Student project repositories
- Database servers
- Academic documentation
- Online meeting requirements
- Graphics and project-design requirements

The existing software environment is based mainly on proprietary solutions..

B. Existing Software Environments

The representative organization may have a mixed software-development ecosystem consisting of local development environments, source-code directories, project documentation, issue-management processes, databases, application servers and manually coordinated deployment procedures.

Software	Purpose
Microsoft Office	Documents, spreadsheets and presentations
GitHub Enterprise	Source-code hosting and project collaboration
Microsoft SQL Server	Database management
Adobe Photoshop	Image editing and graphics
Zoom	Online meetings and communication

The major concern is the combined cost of maintaining multiple proprietary software products.

III. PROBLEM ANALYSIS

A. Existing Challenges

The college may face the following challenges:

1. Licensing Cost : Multiple proprietary applications may require paid licenses or subscriptions.

2. Vendor Dependency : The college depends on individual vendors for software updates, features and service availability.

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3. Limited Student Access : Students may only be able to use licensed software through college systems.

4. Compatibility : Moving between different software platforms can create file-format and application compatibility issues.

5. Training : Students and faculty already familiar with existing software may require training after migration.

6. Data Migration : Existing documents, databases and project repositories must be migrated carefully.

7. Specialized Features : Some proprietary applications may provide features that are not exactly identical in their open-source alternatives.

IV. OPEN SOURCE SOLUTION ANALYSIS

A. Proposed Open-Source Alternatives

The proposed technology stack is:

Existing Software	Proposed Alternative	Main Purpose
Microsoft Office	LibreOffice	Office productivity
GitHub Enterprise	GitLab Community Edition	Git repository and collaboration
Microsoft SQL Server	PostgreSQL	Database management
Adobe Photoshop	GIMP	Image editing
Zoom	Jitsi Meet	Video conferencing
Proprietary OS where applicable	Linux	Operating system

*B. Microsoft Office → LibreOffice*Git

Existing System
Microsoft Office is used for:

- Assignments
- Reports
- Presentations
- Spreadsheets
- Academic documentation

Proposed Solution

LibreOffice can be introduced as the open-source office suite.

Its major components include:

- Writer – word processing
- Calc – spreadsheets
- Impress – presentations

Benefits

- Open-source office suite
- Supports common academic tasks
- Supports common document formats
- Suitable for student laboratories
- Reduces dependence on proprietary office software

Migration Concern

Existing documents should be tested before complete migration because complex formatting, macros or proprietary features may not always behave identically.

C. GitHub Enterprise → GitLab Community Edition

Existing System

- GitHub Enterprise is used for:
- Source-code repositories
- Student project collaboration
- Version control
- Code review

Proposed Solution

GitLab Community Edition can provide a self-hosted open-source platform for Git-based development and collaboration.

It can support:

- Git repositories
- Branches
- Merge requests
- Issue tracking
- Project management
- CI/CD capabilities

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Benefits :

- Self-hosted deployment possible
- Greater institutional control
- Open-source platform
- Suitable for software-development projects
- Can integrate with existing Git repositories

Migration Concern:

Existing repositories, users, permissions, issues and project workflows must be migrated and tested.

D. SQL Server → PostgreSQL

Existing System :
Microsoft SQL Server is used to store and manage structured project and academic data.

Proposed Solution :

PostgreSQL can be used as the open-source relational database system.

It supports:

- Relational data
- SQL
- Transactions
- Constraints
- Indexing
- Database administration
- Application integration

Migration Concern :

Database schemas, stored procedures, functions, queries and data types should be analysed before migration.
Some SQL Server-specific features may require modification.

E. Adobe Photoshop → GIMP

Existing System :

Adobe Photoshop is used for:

- Image editing
- Project graphics
- Posters
- Image manipulation
- Documentation graphics

Proposed Solution :

GIMP (GNU Image Manipulation Program) can be used for many common image-editing requirements.

Benefits:

- Open-source
- Suitable for common image-editing tasks
- No commercial Photoshop license required
- Useful for student projects and graphics work

Migration Concern :

Photoshop-specific files, plugins and advanced workflows may require additional testing or alternative methods.

F. Zoom → Jitsi Meet

Existing System:

Zoom is used for:

- Online classes
- Meetings
- Project discussions
- Faculty-student communication

Proposed Solution:

Jitsi Meet can be considered as an open-source video-conferencing alternative.

It provides features such as:

- Video meetings
- Audio communication
- Screen sharing
- Meeting-based collaboration

For institutional deployment, the college can evaluate self-hosting requirements, server capacity and network performance.

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V. SOFTWARE COMPARISON

A. MS Office vs. LibreOffice

Parameter	MS Office	LibreOffice
Model	Proprietary	Open Source
Main Use	Office productivity	Office productivity
Documents	Yes	Yes
Spreadsheets	Yes	Yes
Presentations	Yes	Yes
License Cost	Paid licensing/subscription depending on edition	No proprietary license fee
Compatibility	Strong with Microsoft formats	Supports common formats
Migration Difficulty	Existing environment	Low–Medium

B. GitHubb vs. GitLab

Parameter	GitHub Enterprise	GitLab CE
Model	Proprietary enterprise platform	Open-source/self-hostable edition
Repository	Yes	Yes
Collaboration	Yes	Yes
Issue Tracking	Yes	Yes
CI/CD	Yes	Yes
Self Hosting	Enterprise deployment options	Yes
Migration	Existing system	Requires repository/user migration

C. SQL Server vs PostgreSQL

Parameter	SQL Server	PostgreSQL
Model	Proprietary	Open Source
Database Type	Relational	Relational
SQL Support	Yes	Yes
Transactions	Yes	Yes
Scalability	Yes	Yes
Migration	Existing database	Requires schema/data conversion
License Model	Commercial editions/licenses	Open-source license

D. Photoshop vs. GIMP

Parameter	Photoshop	GIMP
Model	Proprietary	Open Source
Image Editing	Yes	Yes
Layers and Filters	Yes	Yes
Advanced Professional Features	Extensive	Different feature set
Cost Model	Paid	Open Source
Migration	Existing environment	Requires user adaptation

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E. Zoom vs. Jitsi Meet

Parameter	Zoom	Jitsi Meet
Model	Proprietary	Open Source
Video Meetings	Yes	Yes
Audio	Yes	Yes
Screen Sharing	Yes	Yes
Self Hosting	Not the same model as OSS self-hosting	Possible
Cost	Subscription plans available	Software itself can be deployed without proprietary license fees
Migration	Existing users	Requires user training/configuration

VI. COST-BENEFIT ANALYSIS

A. Existing Cost Categories

The existing system may involve costs for:

- Software licenses
- Subscription renewals
- Enterprise collaboration software
- Database licensing
- Graphics software
- Video-conferencing plans
- Vendor support
- Software upgrades

B. Proposed OSS Cost Categories

Open-source software can reduce proprietary license fees, but the college should still consider:

- Server hardware
- Cloud infrastructure
- System administration
- Technical support
- Training
- Data migration
- Backup infrastructure
- Security monitoring

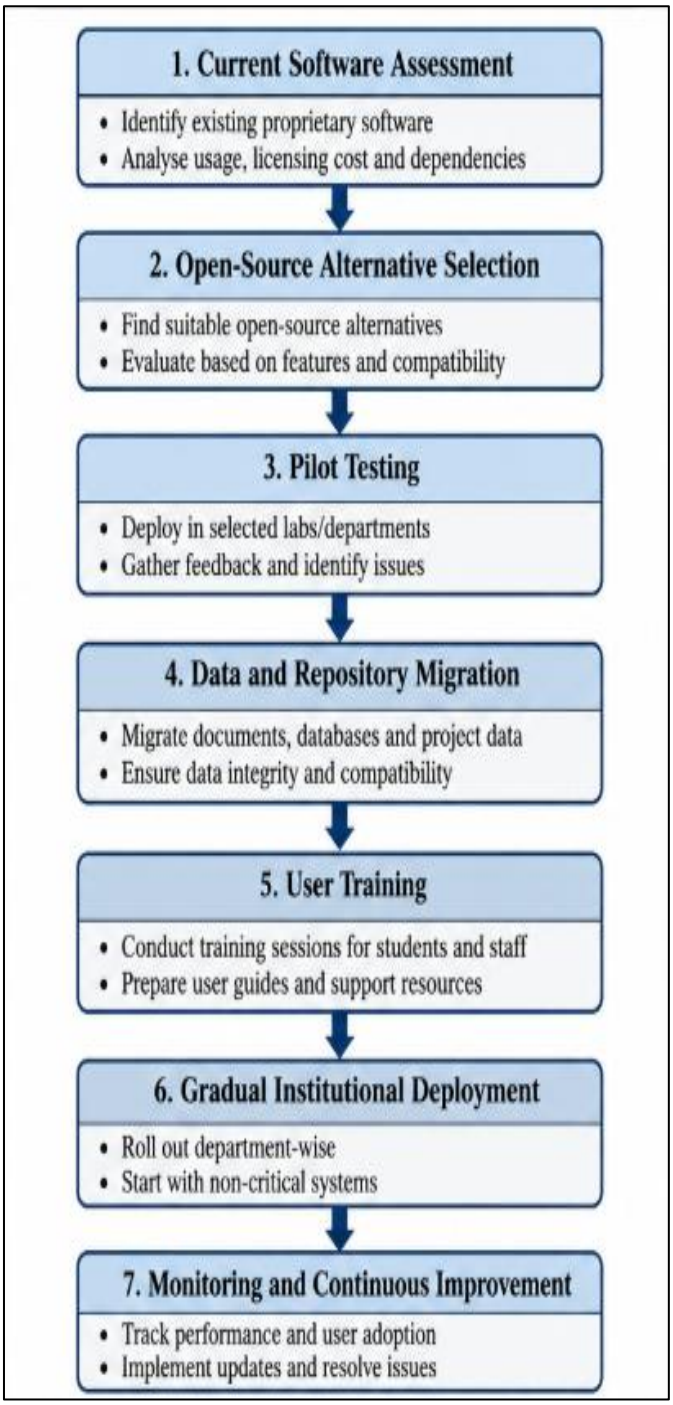
Therefore : Open source does not necessarily mean zero cost. It mainly changes the cost structure from software licensing toward infrastructure, training, deployment and maintenance.

Cost-Benefit Table

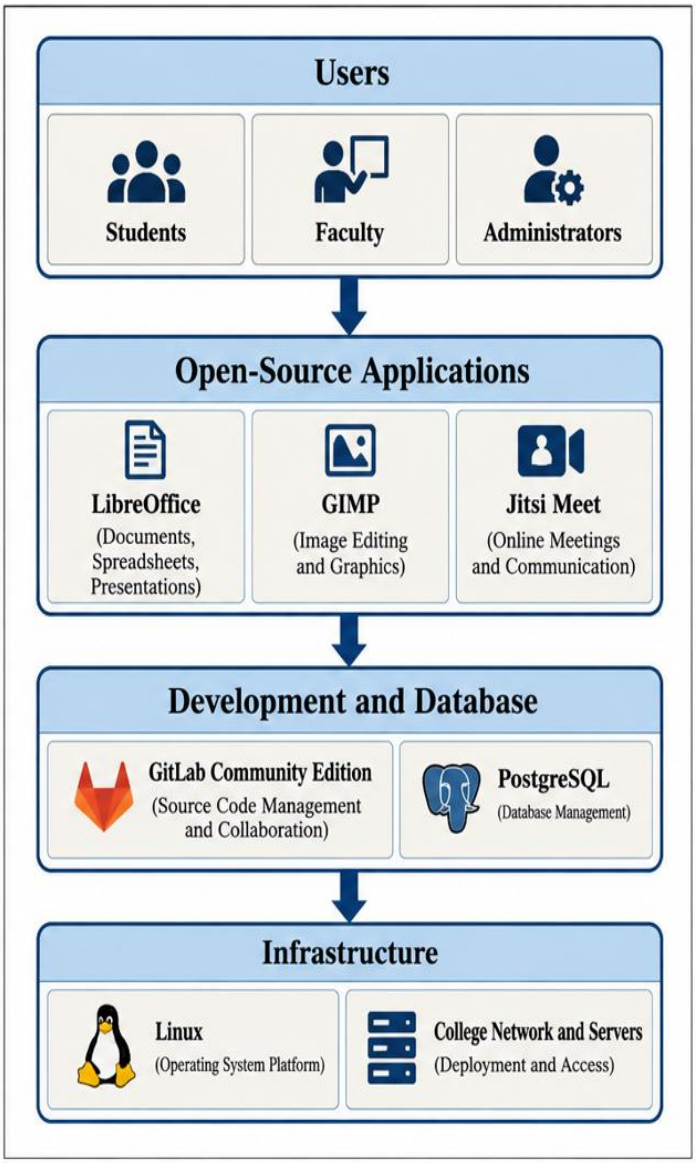
Factor	Existing Proprietary System	Proposed OSS System
Software Licensing	Recurring/paid	Reduced proprietary licensing
Vendor Dependency	Higher	Lower
Initial Migration Cost	Low	Medium
Training Cost	Existing users familiar	Initial training required
Infrastructure Cost	Depends on product	May increase for self-hosted systems
Customization	Vendor/platform dependent	Greater flexibility
Maintenance	Vendor + internal	Mainly institutional/technical team
Student Accessibility	Depends on licenses	Generally easier to deploy
Long-Term Flexibility	Vendor dependent	Greater control

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VII. OSS MIGRATION ROADMAP



VIII. SYSTEM ARCHITECTURE



IX. SECURITY AND LICENSING

Open-source migration must also consider security.

1) *Security Measures*

- Regular security updates
- Strong user authentication
- Role-based access
- Firewall configuration
- Regular backups
- Secure database configuration
- Repository access control
- Monitoring and logging

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- Restricted administrator privileges

2) *Licensing*

The college should maintain a software inventory and verify the license of each open-source component.

Software	License/Model
LibreOffice	MPL 2.0
GIMP	GPL
PostgreSQL	PostgreSQL License
GitLab CE	Open-source licensing for the applicable distribution/version
Jitsi	Open-source project with applicable component licenses
Linux	Depends on distribution and included components

X. CHALLENGES AND RISK ANALYSIS

Challenge	Impact	Mitigation
User resistance	Medium	Training and demonstrations
File compatibility	Medium	Pilot testing and conversion
Database migration	High	Backup and staged migration
Repository migration	Medium	Test migration before final transfer
Photoshop feature differences	Medium	Training and workflow testing
Video conferencing performance	Medium	Network/server testing
Lack of technical skills	Medium	IT staff training
Security configuration	High	Regular updates and monitoring

XI. EXPECTED BENEFITS

The proposed migration can provide the college with:

1) Reduced Proprietary Software Dependency
The college becomes less dependent on individual commercial vendors.

2) Better Software Accessibility
Open-source applications can be deployed across suitable systems without purchasing individual proprietary licenses.

3) Student Learning
Students gain practical experience with:
- Linux
- Git
- GitLab
- PostgreSQL
- LibreOffice
- GIMP
- Open-source collaboration

4) Greater Flexibility
The institution has greater flexibility to configure and integrate software according to its requirements.

5) Long-Term Cost Management
Although migration has initial costs, reducing recurring proprietary license expenses can change the institution's long-term software expenditure.

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XII. RECOMMENDATIONS

The following recommendations are proposed:

1. Do not migrate all software simultaneously.
2. Start with a pilot laboratory.
3. Perform complete software and hardware assessment.
4. Back up all important data before migration.
5. Test file and database compatibility.
6. Train faculty and students before deployment.
7. Establish an internal technical support team.
8. Maintain an open-source software inventory.
9. Apply security updates regularly.
10. Keep proprietary software temporarily where no suitable alternative exists.
11. Review the migration results periodically.
12. Use user feedback to improve the migration process.

XIII. CONCLUSION

ABC Engineering College currently depends on several proprietary software solutions for academic and project-related activities. The increasing cost of licensing provides a reason to evaluate open-source alternatives.

This case study proposes LibreOffice, GitLab Community Edition, PostgreSQL, GIMP and Jitsi Meet as alternatives for Microsoft Office, GitHub Enterprise, SQL Server, Adobe Photoshop and Zoom respectively. Linux can additionally provide an open-source operating-system platform for suitable college systems.

The migration should not be treated as a simple software replacement. It requires assessment, compatibility testing, data migration, training, security planning and continuous support. A phased migration strategy allows the college to test the proposed solutions before wider deployment.

The expected result is an environment with reduced dependence on proprietary software, greater exposure to open-source technologies and improved flexibility in managing the institution's software infrastructure.

XIV. REFERENCES

[1] Open Source Initiative, “The Open Source Definition,” Open Source Initiative. https://opensource.org/osd

[2] The Document Foundation, “LibreOffice Licenses,” LibreOffice. https://www.libreoffice.org/licenses/

[3] GitLab Inc., “GitLab Community Edition,” GitLab. GitLab Community Edition Documentation

[4] PostgreSQL Global Development Group, “PostgreSQL License,” PostgreSQL. https://www.postgresql.org/about/licence/

[5] PostgreSQL Global Development Group, “PostgreSQL Documentation,” PostgreSQL. https://www.postgresql.org/docs/

[6] The GIMP Development Team, “GIMP Frequently Asked Questions,” GIMP. https://www.gimp.org/docs/userfaq.html

[7] Jitsi, “About Jitsi Meet,” Jitsi. https://jitsi.org/jitsi-meet/

[8] Open Source Initiative, “Frequently Answered Questions,” Open Source Initiative. https://opensource.org/faq